

PREDICTING FINANCIAL DISTRESS FROM ORBIS DATA: a machine learning approach

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STUDY PURPOSE

Understanding the drivers of firms' financial distress

Exploiting Orbis (BvD) large dataset to extend the existing literature on predicting business default (Altman, 1968 and Altman, 1983)

DATA IMPORTATION

- Sampling strategy: random draw for 48000 active and distressed firms (evenly split)
- Distress → firm is insolvent/bankrupt/liquidated
- Some requisites imposed. Firms: Medium to Very Large companies. Financial history: 3–10 years available
- Variables required: Assets, Sales, EBIT, Working Capital, Retained Earnings, Equity, Debt
- Excluded: Public entities, firms with missing recent data (no prediction possible for investors)

NA TREATMENT

Variables with excessive missingness were removed (not useful):

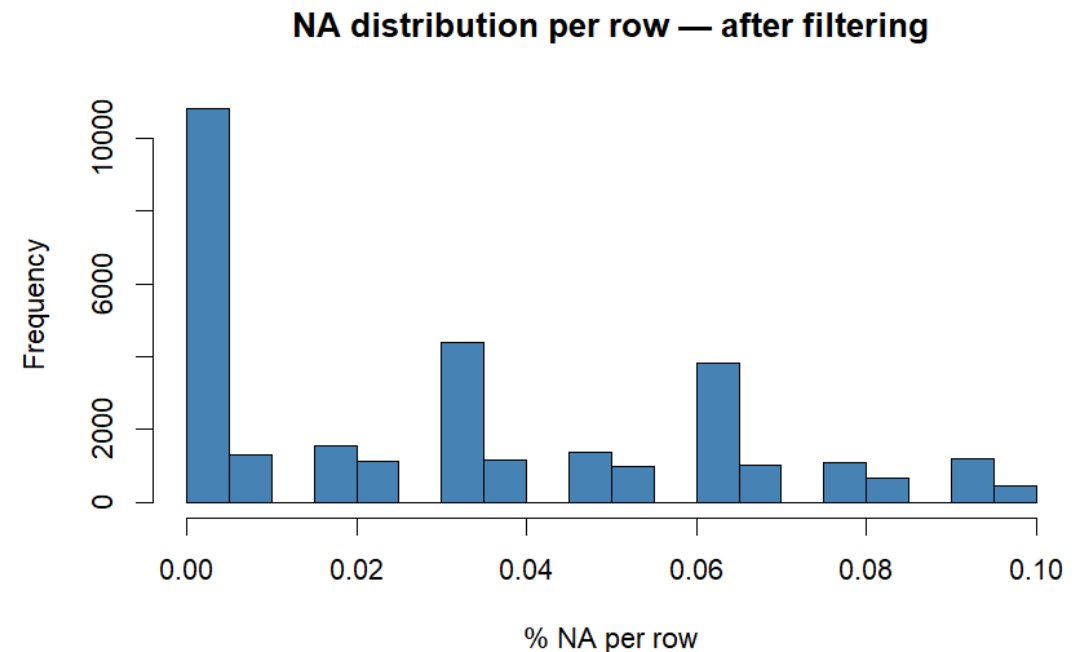
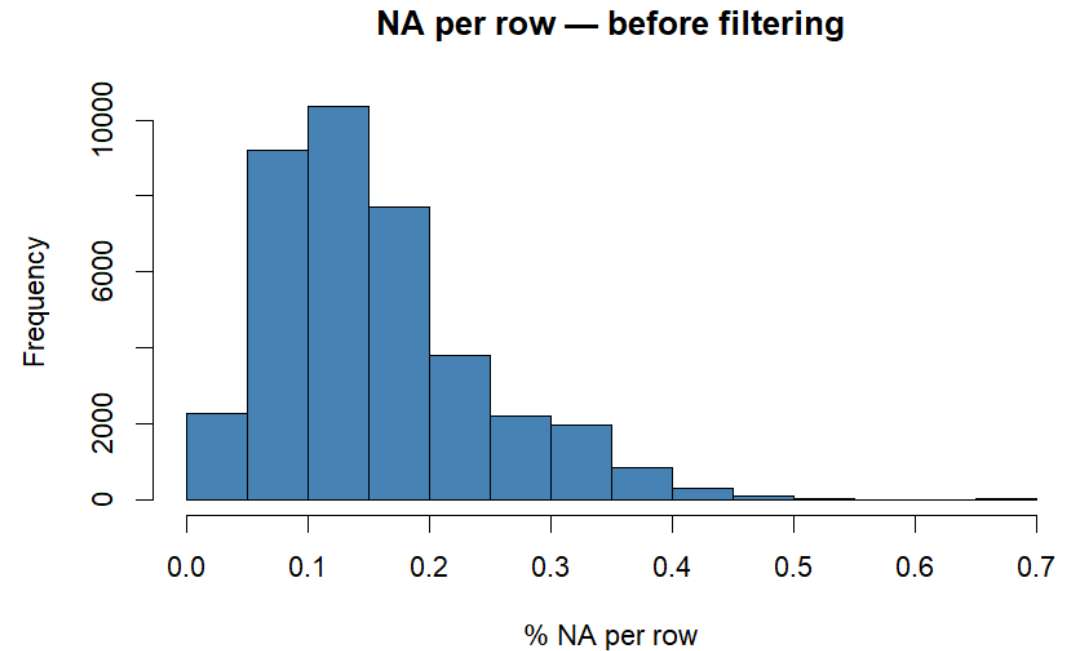
- Variables removed with more than 50% missing values: **8**
- Additional variables removed with more than 25% missing values: **28**

Rows with excessive missingness were excluded (not useful but possible bias):

- First threshold: maximum **20% NA per observation**
- Final threshold: maximum **10% NA per observation**

After final filtering:

- NA % in the dataset: **3.1%**



Systematic Missingness Assessment

Geographic area

- $\chi^2 = 1164.7, p < 0.001$
- Firm size
- $\chi^2 = 73.7, p < 0.001$
- Distress status
- $\chi^2 = 169.96, p < 0.001$

Most affected groups

- South & Central America: **61.3%** observations removed
- Distressed firms: **18.6%** removed

Final Dataset

Final sample size: **30,977 firms**

Non-distressed firms: **17,575**

Distressed firms: **13,402**

Class ratio:

approximately **1 distressed firm**
every 1.3 active firms

FEATURE ENGINEERING

- Firms with non-positive total assets or liabilities excluded
- Firm age constructed from foundation year and fiscal year
- Financial ratios and balance-sheet indicators computed from raw accounting variables
- Outliers treated through winsorization (1st–99th percentile)
- Temporal variation features generated up to three years before
- **Dynamic features included to capture financial deterioration and trend effects preceding distress**

TRAIN/TEST SPLIT AND IMPUTATION

- Stratified train/test split: 70/30, class proportions preserved across datasets
- Missing-value imputation performed after split to avoid data leakage, imputation parameters estimated exclusively on training
- Imputation via mean + Gaussian noise (other approaches were tested such as KNN imputation)

K MEANS CLUSTERING

	Cluster 1	Cluster 2
Active firms	12046	20
Distress	9322	2

Pearson's Chi-squared test

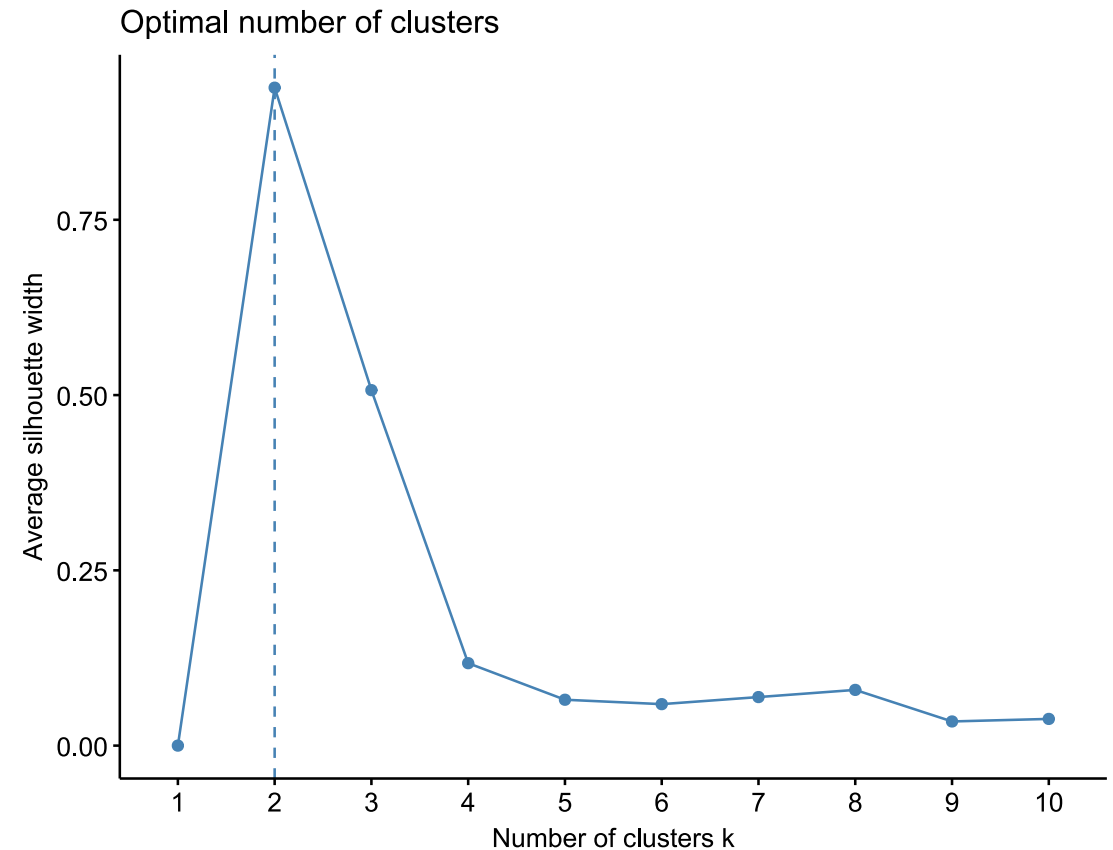
X-squared = 9.3017, df = 1, p-value < 0.002289

K means identify a small cluster of very large multinationals. We consider them outliers and rerun the algorithm without them

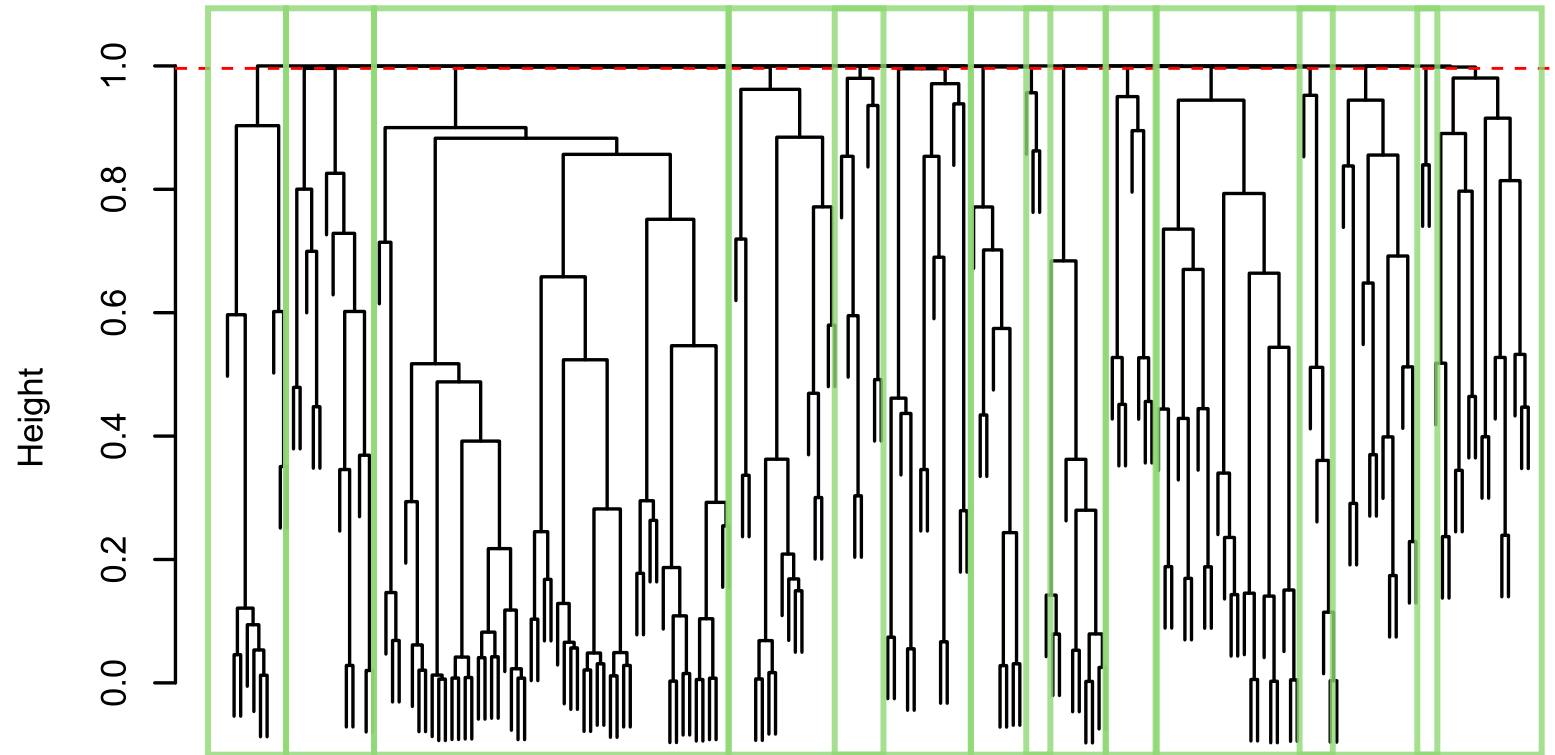
	Cluster 1	Cluster 2
Active firms	11597	449
Distress	9297	25

Pearson's Chi-squared test

X-squared = 288.32, df = 1, p-value < 2.2×10^{-16}



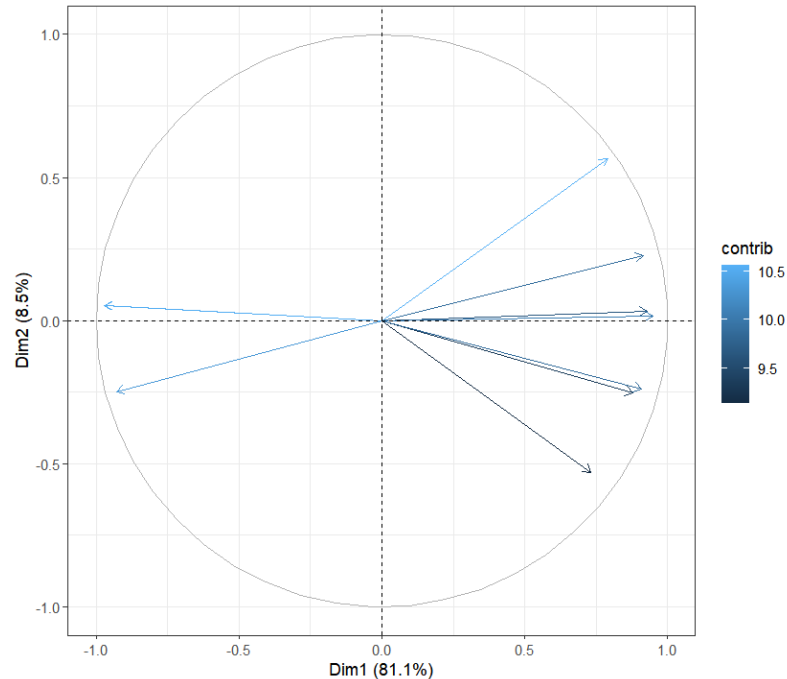
H. CLUSTERING OF VARIABLES - COMPLETE



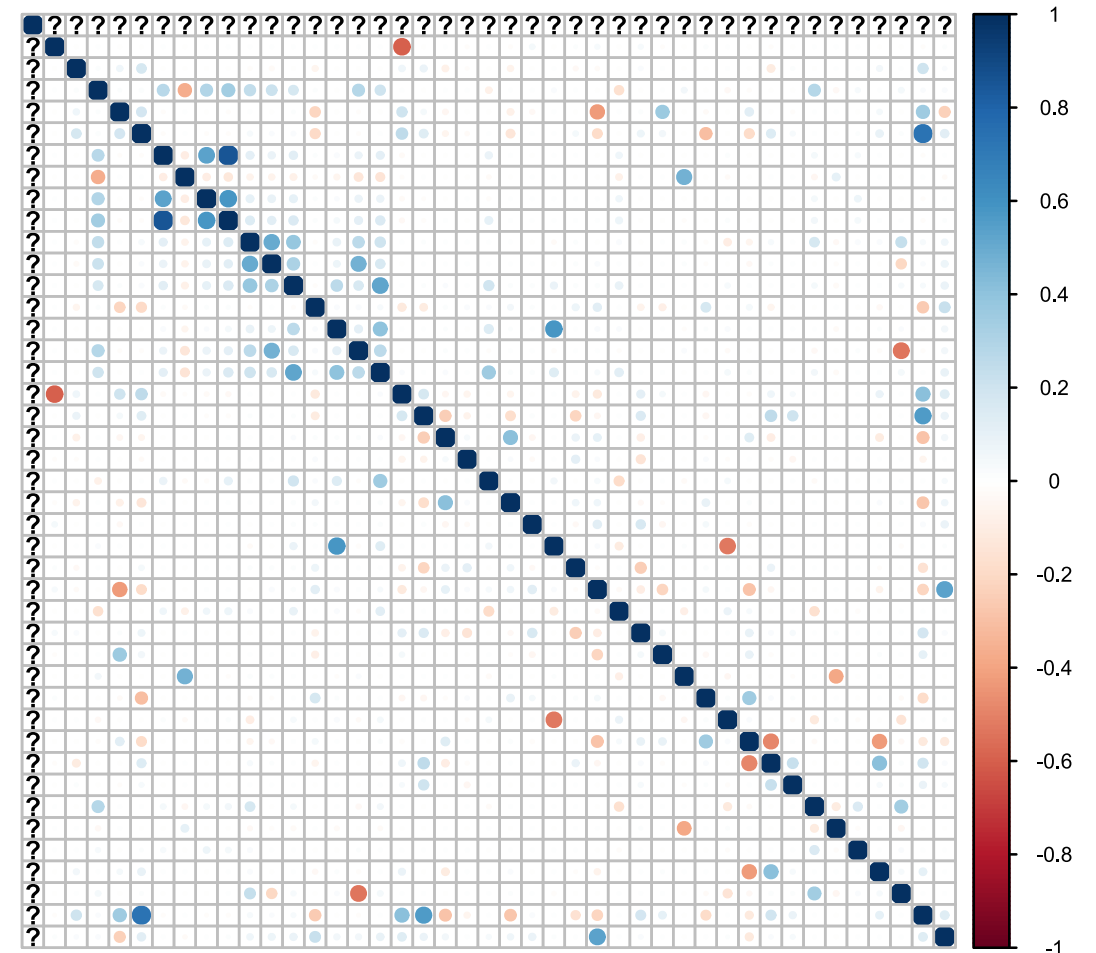
Euclidean distance
computed on the
absolute value of the
correlation
coefficient r

We select 15 clusters of height 0.996003, so that the variables of the same group can be more or less correlated, but between-groups correlation is $|r| < 0.003997$, i.e. variables belonging to different groups are practically uncorrelated.

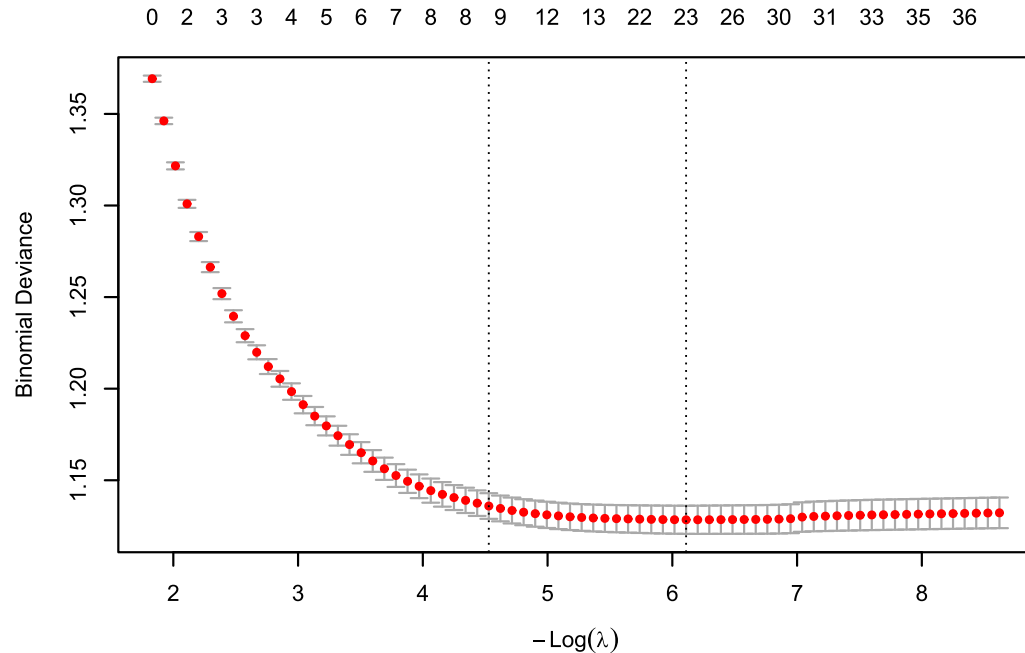
SELECTING VARIABLES BASED ON LOADINGS



For each cluster, we choose the variable with the highest loading over each PC up to 70% explained variance. We end up with 49 variables for a total of 50 quantitative variables (adding the age of the firm)

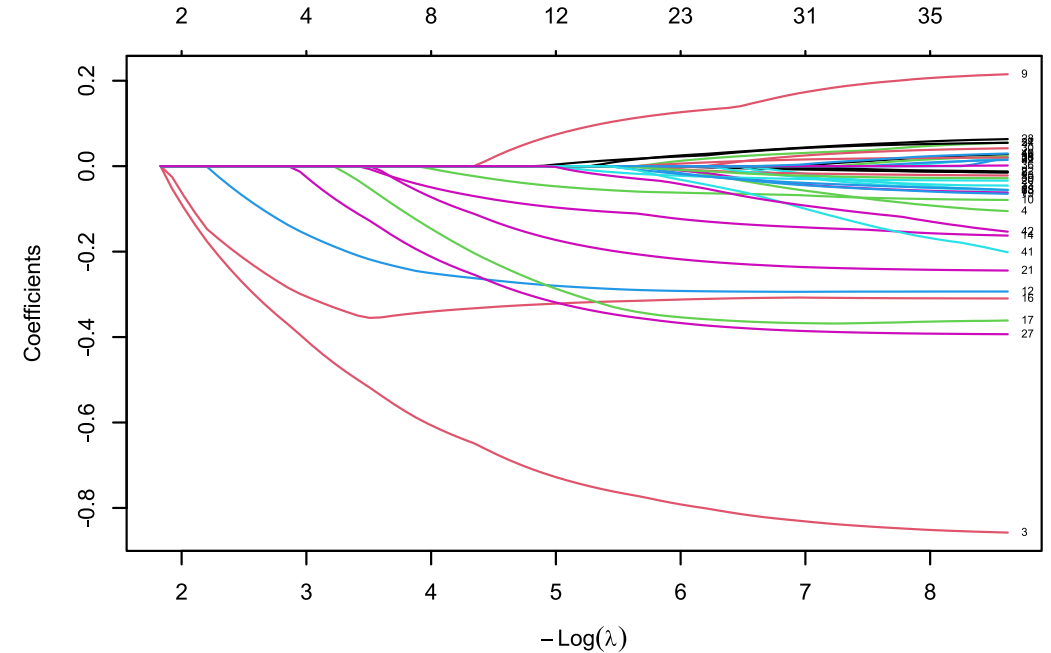


LASSO BINOMIAL REG.



We can safely keep just 9 quantitative variables

After the LASSO, all VIFs are below 5, so no relevant multicollinearity problems emerge.



Tracing the most important variables:

1. Coefficiente di solvibilità (sulla base del patrimonio) Anno – 2
2. Redditività del totale Attivo (ROA) - Lordo Ultimo anno disp.
3. Margine EBIT Ultimo anno disp.
4. sub_T2_Coefficiente di solvibilità (sulla base del patrimonio) Anno - 1

LASSO-DRIVEN LOGISTIC REGRESSION

	<i>Dependent variable:</i>
	Distress
Solvency Ratio (t-2)	-0.026***
Liquidity Ratio	0.059***
EBIT Margin (t-2)	-0.007***
EBIT Margin	-0.012***
ROE (Gross)	-0.002***
ROA (Gross)	-0.020***
Sales Revenue (t-2)	-0.000002***
Solvency Ratio (t)	-0.009***
Solvency Ratio (t-1)	-0.022***
Constant	-2.584**

Note: *p<0.1; **p<0.05; ***p<0.01

	<i>Dependent variable:</i>
	Distress
Solvency Ratio (t-2)	-0.0219***
EBIT Margin (t-2)	-0.0154***
Sales Revenue (t-2) (thousands)	-0.00000217***
Constant	-2.629**

Note: *p<0.1; **p<0.05; ***p<0.01

Both include geographical and temporal effects.

OOS Classification: ALL v. LASSO v. ONLY t-2

ALL selected variables	
Accuracy	84.4%
p-value	$<2.2 \times 10^{-16}$
Sensitivity	75.83%
Specificity	90.89%
Kappa	67.72%

LASSO-driven (thr.=0.5)	
Accuracy	84.12%
p-value	$<2.2 \times 10^{-16}$
Sensitivity	76.16%
Specificity	90.16%
Kappa	67.2%

ONLY t-2 (thr.=0.4)	
Accuracy	79.67%
p-value	$<2.2 \times 10^{-16}$
Sensitivity	80.18%
Specificity	79.29%
Kappa	67.72%

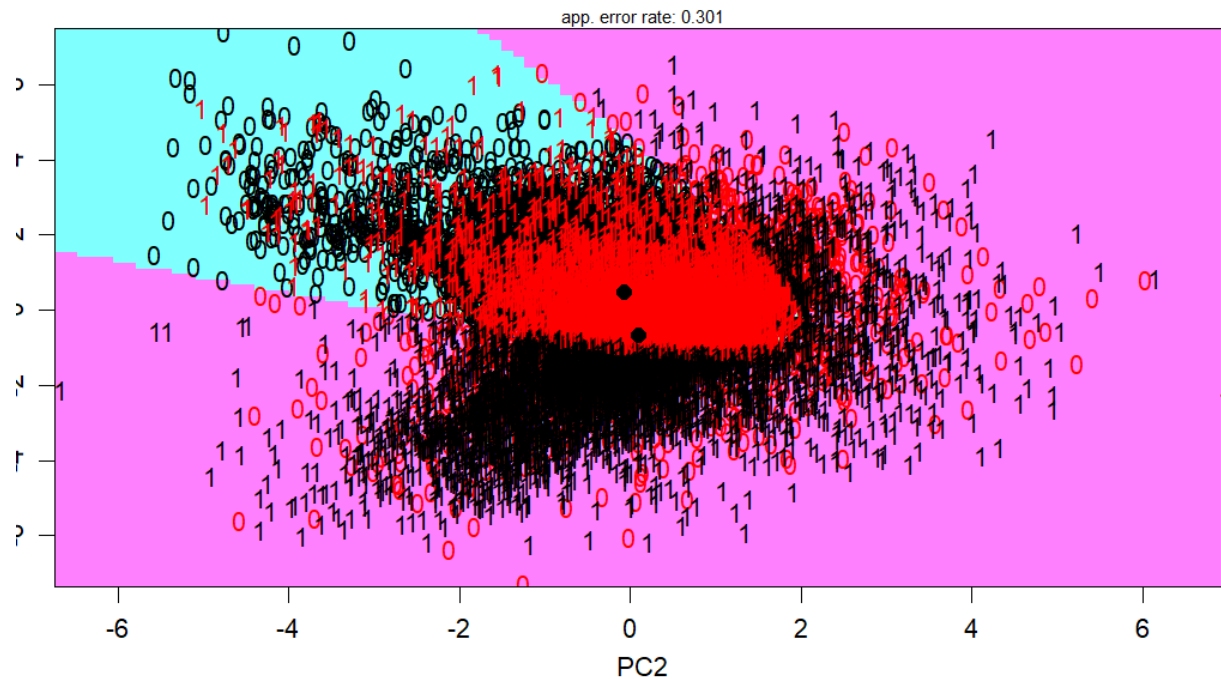
LASSO-driven (FT, thr.=0.4)	
Accuracy	83.44%
p-value	$<2.2 \times 10^{-16}$
Sensitivity	82.76%
Specificity	83.95%
Kappa	66.4%

t-2, without YEAR (thr.=0.5)	
Accuracy	71.87%
p-value	$<2.2 \times 10^{-16}$
Sensitivity	71.49%
Specificity	72.17%
Kappa	43.29%

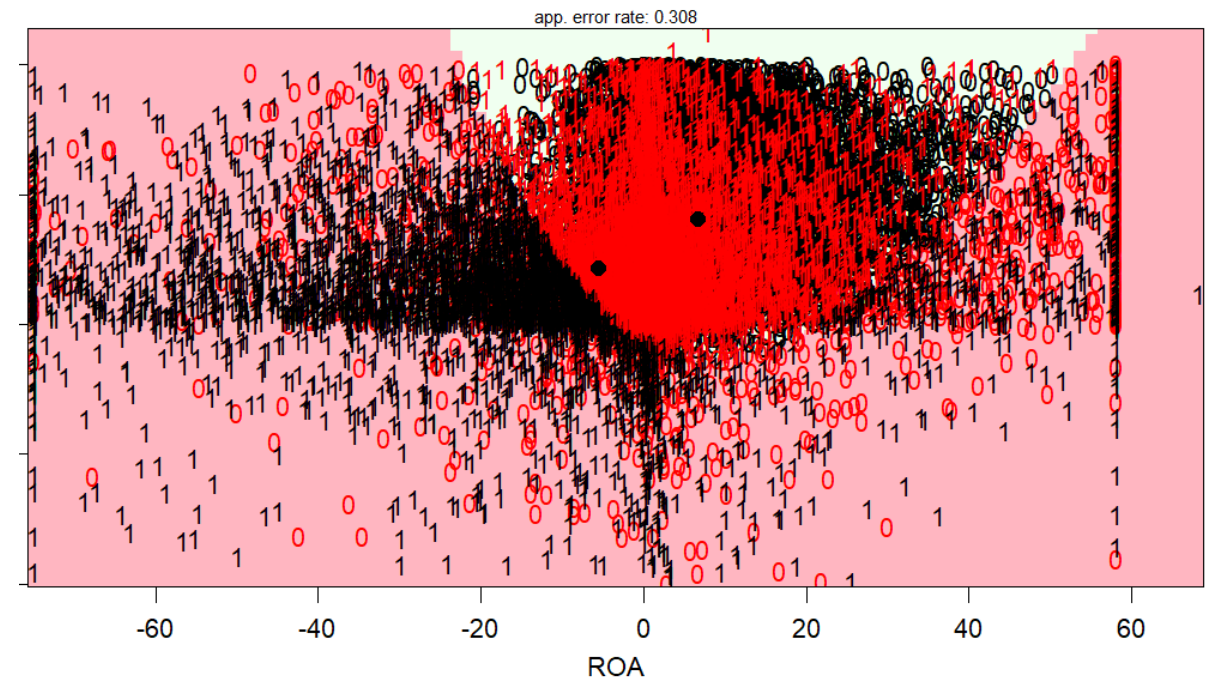
Does the classification emerge from a simple partition plot?

QDA chosen for two reasons: 1) high sample dimension; 2) frail assumption of equal v-cov between classes

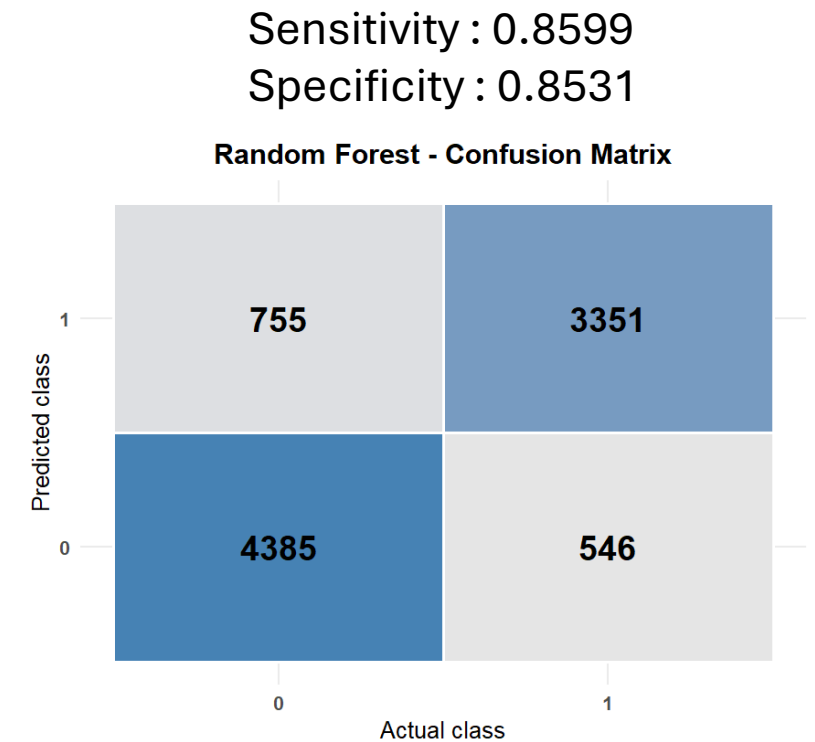
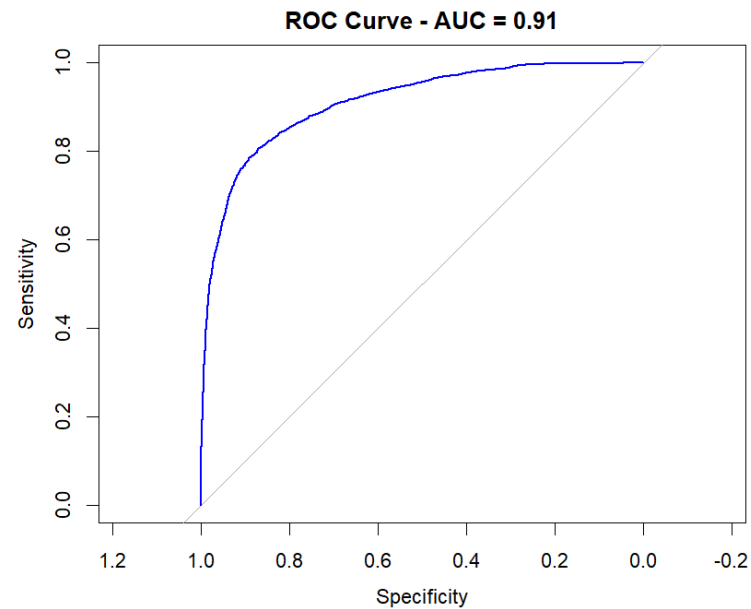
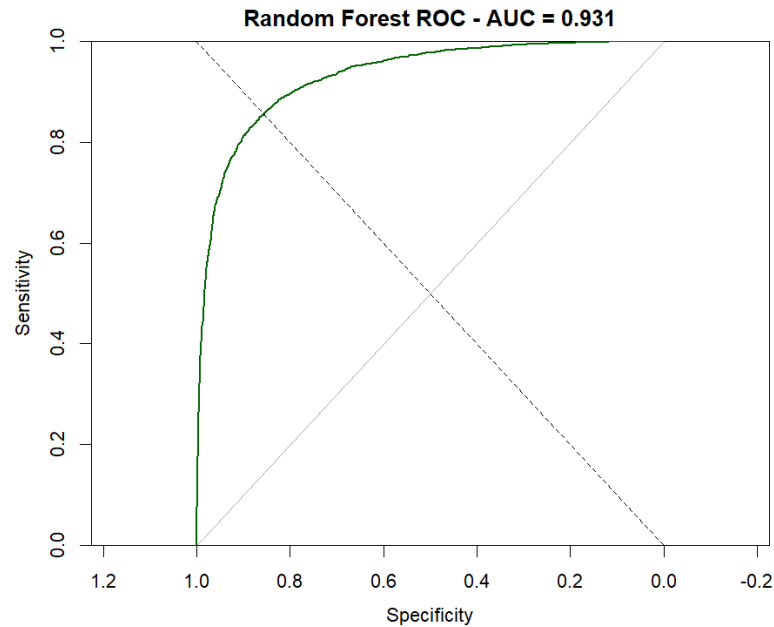
Partition Plot



Decision Boundaries: QDA



A non-parametric approach performs better?



Conclusions

We built a coherent methodology for data cleaning and the screening of the feature space bridging different statistical learning techniques.

It is possible to design a data-driven model to reliably predict financial distress.

Moreover, a LASSO-driven selection finds few very important explanatory variables with negligible loss in predictive power.

The variables employed reflect core entrepreneurial dimensions of redditivity, solvency, and liquidity. Moreover, geographical and temporal effects are relevant.

Limitations: The dataset had a great amount of missing data, and it was systematically cleaned. This process might produce some bias; however, this does not emerge from Out Of Sample predictions.

References

- Orbis Europe: <https://orbis-r1.bvdinfo.com/version-20260316-5-2/Orbis/1/Companies/Search>;
- Code and Replication material: <https://github.com/paolochiabrera/Financial--distress>;
- Altman (1968) and (1983);
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning* (1st ed.) [PDF]. Springer;
- Horton NJ, Kleinman KP. (2007). “Much ado about nothing: A comparison of missing data methods and software to fit incomplete data regression models.” Am Stat.